

Bayesian optimization approach for agent-based modeling of cancer cell migration and invasion phenotype

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Abstract

Collective cancer cell invasion is a defining hallmark of metastatic progression, characterized by coordinated migration and dynamic transitions between noninvasive spheroid and invasive network phenotypes. These emergent behaviors arise from complex, nonlinear interactions among biophysical and biochemical processes, including cell–cell adhesion, chemoattractant secretion, and extracellular matrix (ECM) remodeling. Although agent-based and multiscale frameworks such as the cellular Potts model implemented in CompuCell3D provide powerful mechanistic insight into these processes, their predictive capability critically depends on accurate, reproducible parameterization. In practice, parameter estimation remains challenging due to high-dimensional search spaces, stochastic simulation dynamics, and substantial computational cost.

Here, we present a data-driven, constrained Bayesian optimization (BO) framework for efficient calibration of agent-based models of collective cancer cell invasion against experimental observations. Quantitative phenotypic metrics, including cell invasion distance and morphological cell cluster circularity, are used to define objective functions that capture discrepancies between simulated and experimental data. A Gaussian process (GP) surrogate model, coupled with a feasibility-informed acquisition (FIA) strategy, enables efficient exploration of the parameter space while enforcing biologically grounded constraints. Importantly, the FIA formulation constitutes a physics-informed machine learning approach, in which domain-specific knowledge, physical and morphological characteristics of cell organization, such as circularity, are explicitly incorporated into the optimization process as

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constraints. This integration ensures that candidate solutions are not only optimal with respect to the objective function but also remain consistent with experimentally observed biophysical behavior. Subsequent Bayesian inference further refines parameter estimates, yielding posterior distributions and principled uncertainty quantification.

Applied to spheroid-to-network invasion transitions, the proposed framework achieves strong agreement between *in silico* and *in vitro* phenotypes while revealing key mechanistic trade-offs among adhesion strength, chemotactic signaling, and chemoattractant secretion dynamics. In particular, we demonstrate that observed morphological phenotypes can be accurately recapitulated by minimizing a weighted root mean square error (WRMSE) with respect to experimental invasion measurements under explicit circularity constraints. This integrated approach establishes a robust and scalable methodology for data-calibrated agent-based modeling and advances the development of predictive, uncertainty-aware computational tools for cancer systems biology.

Keywords: Agent-based model (ABM), collective cell migration, Bayesian inference, Bayesian optimization, data-driven parameter estimation, surrogate model, feasibility-informed acquisition function, cell phenotypes

1. Introduction

Collective cancer cell invasion represents a fundamental hallmark of cancer for metastatic progression, in which groups of cancer cells migrate cohesively while maintaining cell-cell adhesion and structural coordination¹. This mode of invasion contrasts with single-cell migration and is often characterized by dynamic transitions between spheroid clusters and network phenotypes morphologies^{2–4}. Such phenotypic plasticity is critical to cancer dissemination and therapeutic resistance^{5,6}. Understanding the underlying mechanisms that drive this transition requires quantitative models capable of linking cell-level biophysical and biochemical interactions to emergent, tissue-scale invasion patterns.

Agent-based modeling (ABM) within multiscale computational frameworks, such as the cellular Potts model (CPM) implemented in CompuCell3D (CC3D)⁷, has become a powerful tool for simulating complex cellular processes. However, predictive capability depends heavily on accurate parameterization. Parameters governing cell-cell and cell-matrix adhesion energies, chemoattractant secretion rates, and mechanical constraints collectively shape the emergent invasion phenotype⁸. Hence, capturing experimentally observed transitions from spheroid to network morphologies requires a robust, data-informed parameter estimation strategy.

Despite the flexibility of computational cell-based models, identifying optimal parameter sets that accurately reproduce experimental behaviors remains a significant challenge. Parameter spaces in these models are typically multidimensional, nonlinear, and computationally expensive to explore⁹. Small changes in parameters can lead to qualitatively

different invasion outcomes, such as spheroids, networks, or dispersed morphologies. Consequently, traditional manual tuning, heuristic trial-and-error, or grid search approaches are inefficient, subjective, and fail to account for parameter uncertainty¹⁰. In modeling collective cancer cell invasion, these difficulties are further amplified by the stochastic nature of cell-cell interactions and the coupling between biochemical and mechanical cues. Achieving agreement between model simulations and experimental data, therefore, requires systematic approaches that can efficiently navigate complex parameter landscapes while incorporating experimental data as constraints.

Recent advances in probabilistic, data-driven calibration have introduced likelihood-free Bayesian approaches, most notably approximate Bayesian computation (ABC), for parameter inference in complex biological systems^{11–13}. ABC estimates posterior distributions by repeatedly simulating large populations of parameter samples and accepting those that generate outputs sufficiently close to experimental observations¹¹. While effective when likelihoods are intractable, ABC typically relies on population-based sampling, tolerance thresholds, and a large number of forward simulations, making it computationally prohibitive for ABMs of cancer invasion.

Bayesian optimization (BO) (Figure 1) offers a fundamentally different paradigm for parameter estimation in computationally expensive simulation models¹⁴. Instead of sampling full parameter populations, BO constructs a surrogate model, commonly via Gaussian process (GP) regression, that approximates the relationship between the ABM parameters to be estimated and a scalar objective function^{10,12,13,15–18}. An acquisition function then sequentially selects new parameter sets that balance exploration of uncertain regions of the GP with exploitation of promising solutions for the optimal value of the GP as a function of the ABM parameters, enabling sample-efficient optimization with substantially fewer simulations than would be required for ABC. Additionally, using a GP as the surrogate model for the objective function and the constraint yields a tractable likelihood functions.

Here, we depart from population-based ABC approaches and formulate parameter estimation as a surrogate-driven optimization problem using BO. We define a biologically interpretable objective function based on experimentally measured invasion metrics, specifically cell cluster circularity and extent of cell invasion over time from Chen et al.¹⁹, and quantify simulation-experiment mismatch using different loss functions for each metric. The loss functions directly guide the construction and refinement of our surrogate model. Unlike ABC, which approximates posterior distributions through acceptance-rejection over large parameter spaces^{11,20,21}, our framework performs sequential, targeted exploration using a surrogate model informed by the loss metric for invasion. We also incorporate the feasibility of the cell cluster circularity loss function being constrained to inform the acquisition function. This approach avoids repeated population mapping, substantially reduces computational cost, and provides a transparent link between biophysical and biochemical parameters and emergent invasion morphology, while retaining uncertainty-aware optimization. We estimate key parameters, including the rate of chemoattractant secretion and the cell-cell adhesion energy, that govern the transition between the spheroid and network inva-

sion phenotypes. By coupling a mechanistic ABM with surrogate-based BO, our approach enables quantitative, data-consistent parameter inference for our cancer cell invasion model.

2. Methods

2.1. Overview of the calibration workflow

The calibration of mechanistic models to experimental data forms a critical bridge between simulation and reality. To achieve biologically meaningful predictions, the model parameters must be tuned so that *in silico* behaviors replicate experimental observations with quantitative accuracy. In this study, a data-driven calibration framework was designed to systematically align our CC3D simulations of collective cancer cell invasion with *in vitro* observations of emergence of spheroid and network phenotypes from tumor organoids¹⁹. Our CC3D simulations are from a discrete-continuous hybrid model that combines an ABM for cancer cell behavior and a continuous partial differential equation for reaction-diffusion of a secreted chemoattractant chemical. Our calibration framework (Figure 1) incorporates experimental metrics, an objective function based on minimizing the error between the model and experimental results for both phenotypes, and surrogate-assisted optimization to efficiently estimate key parameters governing cell behaviors.

2.2. Experimental metrics and simulation derived outputs

To inform the parameter estimation and model calibration process, we leveraged an experimental dataset reported in the literature by Chen et al.¹⁹. In their study, they obtained time-lapse and endpoint images of cultured cancer cell organoids and reported means and standard errors of the mean across 13 replicates for invasion over 5 time points and across 25 samples of circularity at 2.5 days for each phenotype. Quantitative experimental measurements were extracted from the published figures using WebPlotDigitizer²². We calculated morphological features from our simulation outputs generated in CC3D using image analysis pipelines in ImageJ²³. Specifically, ImageJ was used to calculate circularity metrics from the simulated cell cluster images, enabling direct quantitative comparison between the experimental morphologies reported by¹⁹ and the corresponding simulation-derived phenotypes. The following metrics were computed for both experimental and simulated conditions:

- Invasion (*I*): Invasion distance is quantified as the maximum displacement of cells relative to a common reference point, defined as the initial position of the first cell prior to mitosis. For each cell, the Euclidean distance between its current position and this fixed reference position is computed, and the maximum value across all cells is taken at each time point. This formulation provides a consistent measure of the spatial extent of invasion into the extracellular matrix arising from the underlying biophysical and biochemical interactions in the CC3D framework.

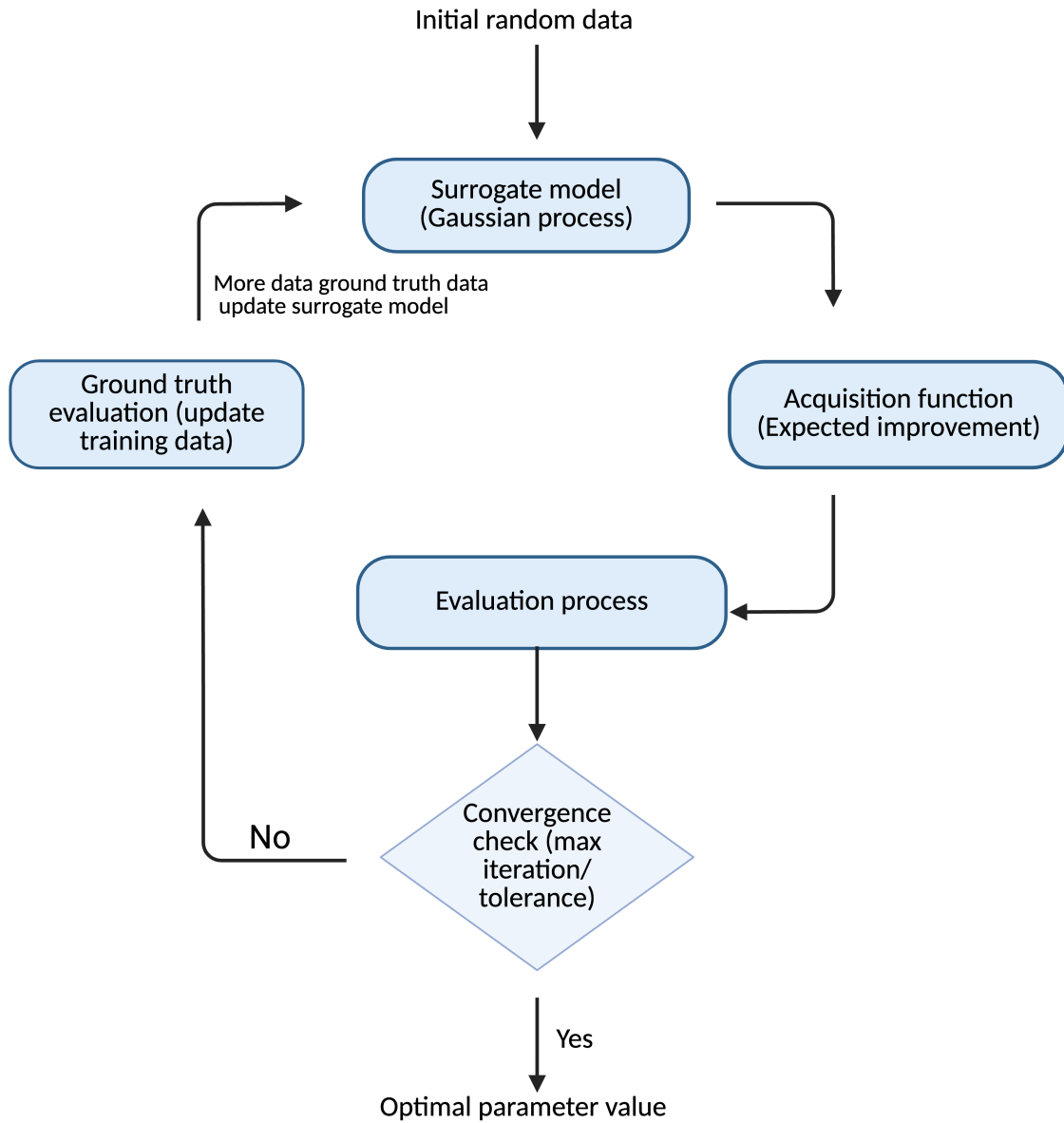


Figure 1: Overview of the surrogate-driven Bayesian optimization framework used to calibrate an agent-based model for cancer invasion. Model parameters are iteratively proposed using an acquisition function applied to a Gaussian process surrogate and a biologically grounded loss function based on experimental metrics, enabling efficient parameter estimation without population-based sampling. Created with BioRender.com.

- **Circularity (C):** Circularity is a metric used to describe the shape of a cell cluster and allows us to quantify morphological changes. Circularity is defined as the ratio of the area of the cell cluster to the area of a circle with the same perimeter. This metric ranges from 0 (indicating the extreme of an elongated network phenotype) to 1 (indicating a perfectly circular spheroid phenotype).

In our model, we have two parameters that comprise the adjustable parameter set: $\theta = \{k, J_{CC}\}$, where k is the chemoattractant secretion rate and J_{CC} is the cell-cell adhesion energy. CC3D simulations were executed for a fixed number of Monte Carlo steps (MCS), typically 7,000, corresponding to the experimental invasion timescale of seven days. From the resulting simulated cellular configurations, we extracted cell invasion $I_{\text{sim}}(\theta)$ and circularity $C_{\text{sim}}(\theta)$ as quantitative metrics. These simulation-derived metrics constitute the observable outputs of the model, providing a functional relationship between the parameter vector θ and experimentally measurable quantities used for model calibration.

2.3. Objective function for Bayesian optimization

We evaluate cell invasion at discrete time points indexed by $i = 1, \dots, N$. Here, $N = 5$. We use phenotype-specific experimental values: $I_{\text{exp},i}^{(p)}$ and $\sigma_{\text{exp},i}^{(p)}$ denoting the experimental mean and standard deviation, respectively, of the invasion metric at time i for phenotype $p \in \{\text{network}, \text{spheroid}\}$. The invasion metric at time i from the average of 20 replicate simulations is denoted $I_{\text{sim},i}$. We quantify the invasion in the simulation as the maximum distance among all cells at time point i to the initial center of the first cell. The cell invasion residual at time i is defined as

$$r_i^{(p)}(I) = I_{\text{sim},i} - I_{\text{exp},i}^{(p)}. \quad (1)$$

Each invasion residual is weighted by the inverse of the experimental variance at that time:

$$w_i^{(p)} = \frac{1}{\left(\sigma_{\text{exp},i}^{(p)}\right)^2}. \quad (2)$$

The weighted root mean squared error for the cell invasion for phenotype p is then given by

$$\text{WRMSE}^{(p)}(I) = \sqrt{\frac{\sum_{i=1}^N w_i^{(p)} \left(r_i^{(p)}\right)^2}{N}}. \quad (3)$$

Let the GP objective function be denoted as L for loss function. Let's consider a few different options:

1. $L = \text{WRMSE}^{(p)}(I)$ from Equation (3)

2. $L = \text{RMSE}_{t_i \geq 1}^{(p)}(I)$ where $i = 1$ starts at time is 1 day and $\text{RMSE}^{(p)} = \sqrt{\frac{\sum_{t_i \geq 1}^N (I_{\text{sim},i} - I_{\text{exp},i}^{(p)})^2}{N}}$

3. $L = \text{RMSE}_{t_i \geq 1}^{(p)}(\Delta I)$ where $\text{RMSE}_{t_i \geq 1}^{(p)}(\Delta I) = \sqrt{\frac{\sum_{t_i \geq 1}^N ((I_{\text{sim},i} - I_{\text{sim},t_i=1}) - (I_{\text{exp},i}^{(p)} - I_{\text{exp},t_i=1}^{(p)}))^2}{N}} = \sqrt{\frac{\sum_{t_i \geq 1}^N (\Delta I_{\text{sim},i} - \Delta I_{\text{exp},i}^{(p)})^2}{N}}$

4. Linear combination of the previous two: $L = \alpha \text{RMSE}_{t_i \geq 1}^{(p)}(I) + (1 - \alpha) \text{RMSE}_{t_i \geq 1}^{(p)}(\Delta I)$

We report circularity mismatch as an absolute error for phenotype p at the final time point:

$$\text{Err}_{\text{circ}}^{(p)} = |C_{\text{sim}} - C_{\text{exp}}^{(p)}|. \quad (4)$$

2.4. Construction of the data-driven surrogate model

Because direct exploration of the parameter space through brute-force simulations is computationally infeasible, a data-driven surrogate model was employed. This surrogate approximates the mapping from model parameters to the objective function, enabling rapid predictions of WRMSE for unseen parameter combinations. The surrogate model $f(\theta)$ was trained on a set of simulation data pairs that was generated using Latin hypercube sampling method:

$$D = \{(\theta_j, \text{WRMSE}_j^{(p)}) \mid j = 1, 2, \dots, N\} \quad (5)$$

where N is the number of completed simulations. Several surrogate modeling strategies could be used for this purpose: GP regression, random forests, or neural networks. We selected GP because it provides smooth function approximation with uncertainty quantification. Random forests capture nonlinear relationships without assuming functional forms, and neural networks are effective for high-dimensional mappings. But both require more training data. Our surrogate model is based on GP because it provides both mean predictions and confidence intervals essential for guiding the BO process. The GP surrogate is trained on the initial simulation runs using Latin hypercube sampling of the parameter space to capture broad parameter-response relationships.

2.5. Bayesian optimization for efficient parameter search

While Bayesian inference provides a principled framework for uncertainty quantification, it can be computationally expensive when each likelihood evaluation requires a full CC3D simulation. In complex biophysical models such as those describing collective cancer cell invasion, the number of parameters and the stochasticity of simulations often make exhaustive sampling infeasible. BO addresses this challenge by combining surrogate modeling with sequential sampling strategies to efficiently search for parameter sets that minimize a target objective function, such as WRMSE, between simulated and experimental cell invasion metrics. Instead of evaluating thousands of random parameter combinations, BO strategically selects new simulation data pairs based on what is most informative, thereby reducing computational cost while maintaining high accuracy. We use BO to identify the optimal combinations of chemoattractant secretion rate and contact energy that best reproduce the experimentally observed transition from spheroid to network invasion phenotypes.

2.6. The Bayesian optimization framework

Bayesian optimization treats the unknown objective function $f(\boldsymbol{\theta})$ (in this case, $f(\boldsymbol{\theta}) = \text{WRMSE}(\boldsymbol{\theta})$) as a random function modeled by a GP surrogate. The GP defines a prior distribution over functions and, after observing data, updates this prior to a posterior that captures both the mean prediction and uncertainty at any point in parameter space. Formally, given an initial set of evaluated points:

$$D_n = \{(\boldsymbol{\theta}_i, f(\boldsymbol{\theta}_i))\}_{i=1}^n, \quad (6)$$

The GP surrogate provides the posterior predictive distribution:

$$f(\boldsymbol{\theta}) \sim \mathcal{N}(\mu_n(\boldsymbol{\theta}), \sigma_n^2(\boldsymbol{\theta})) \quad (7)$$

where $\mu_n(\boldsymbol{\theta})$ is the mean prediction and $\sigma_n^2(\boldsymbol{\theta})$ represents the uncertainty in that prediction. This probabilistic model allows BO to balance *exploration* and *exploitation*. It explores uncertain regions of parameter space while exploiting areas that are likely to yield lower values of the objective function.

2.7. Acquisition function: expected improvement

To select the next parameter set $\boldsymbol{\theta}_{n+1}$ for evaluation, BO uses an acquisition function that quantifies improvements in either reaching optima or in reducing uncertainty by sampling at a given point. One of the most effective acquisition functions, the expected improvement (EI), is defined as

$$EI(\boldsymbol{\theta}) = \mathbb{E}[\max(f^* - f(\boldsymbol{\theta}), 0)] \quad (8)$$

where f^* is the best (minimum) WRMSE observed so far. The EI balances exploration (sampling uncertain regions where $\sigma_n(\boldsymbol{\theta})$ is large) and exploitation (sampling near regions

with low predicted WRMSE). At each iteration, the parameter vector that maximizes EI is recommended for the next CC3D run:

$$\boldsymbol{\theta}_{n+1} = \arg \max_{\boldsymbol{\theta}} EI(\boldsymbol{\theta}) \quad (9)$$

2.8. Constrained Bayesian optimization

To incorporate biologically meaningful constraints, specifically ensuring that simulated spheroid and network morphologies are consistent with experimentally observed circularity, we employed a feasibility-informed acquisition (FIA) strategy²⁴ within the BO framework. The primary objective is to minimize the $f(\boldsymbol{\theta})$, defined as the WRMSE of the maximum cell invasion distance between simulation outputs and experimental observations (Equation (3)). In addition to this objective, we impose a constraint on morphological agreement, defined as

$$g(\boldsymbol{\theta}) \sim \text{Err}_{\text{circ}}^{(p)} \leq g_{\text{threshold}}, \quad (10)$$

where $g(\boldsymbol{\theta})$ is a GP surrogate for the absolute error of the circularity for phenotype p and $\text{Err}_{\text{circ}}^{(p)}$ is defined in Equation (4). To bypass the high computational cost of the simulations, we model both the objective $f(\boldsymbol{\theta})$ and the constraints $g(\boldsymbol{\theta})$ using GP surrogate models. While the GP for the objective function guides the search toward the high feasibility regions, the surrogate for the $g(\boldsymbol{\theta})$ provides a probabilistic estimate of whether a given parameter set $\boldsymbol{\theta}$ will satisfy the morphology constraint. The GP defines a prior distribution over functions and, after observing data, updates this prior to a posterior that captures both the mean prediction and uncertainty at any point in parameter space. Formally, given an initial set of evaluated points:

$$G_n = \{(\boldsymbol{\theta}_i, g(\boldsymbol{\theta}_i))\}_{i=1}^n, \quad (11)$$

The GP surrogate provides the posterior predictive distribution:

$$g(\boldsymbol{\theta}) \sim \mathcal{N}(\mu_n(\boldsymbol{\theta}), \sigma_n^2(\boldsymbol{\theta})) \quad (12)$$

where $\mu_n(\boldsymbol{\theta})$ is the mean prediction and $\sigma_n^2(\boldsymbol{\theta})$ represents the uncertainty in that prediction.

To efficiently navigate the constrained parameter space, we adopt a blended acquisition strategy that balances objective optimization with feasibility exploration. The FIA function is defined as²⁴

$$\alpha_{\text{FIA}}(\boldsymbol{\theta}) = (1 - c_t) \alpha_{\text{EI}}(\boldsymbol{\theta}) + c_t P(\text{feasible} \mid \boldsymbol{\theta}), \quad (13)$$

where $\alpha_{\text{EI}}(\boldsymbol{\theta})$ is the EI based on the objective surrogate and, $P(\text{feasible} \mid \boldsymbol{\theta}) = P(g(\boldsymbol{\theta}) \leq g_{\text{threshold}})$ denotes the probability that the constraint is satisfied at $\boldsymbol{\theta}$, c_t = ratio of infeasible measurements to total measurements, and t is a user-specified hyperparameter (set at value of 1.0), which controls risk when it comes to selecting infeasible measurements. This formulation enables adaptive exploration of feasible regions early in the search while progressively emphasizing objective minimization as feasible solutions are identified, thereby improving both efficiency and robustness in multi-dimensional, stochastic parameter spaces.

2.9. Implementation workflow

The complete constrained BO data-driven calibration workflow was implemented as follows:

1. Initialization: Generate an initial dataset using a combination of Latin hypercube sampling and fixed-grid sampling to ensure broad coverage of the parameter space.
2. Simulation: Execute the CC3D model for each parameter set to obtain phenotypic metrics: maximum cell invasion distance and circularity.
3. Objective and constraint calculation: Compute the objective function WRMSE for cell invasion distance (Equation (3), $f(\boldsymbol{\theta})$) and the constraint function absolute error in circularity (Equation (4), $g(\boldsymbol{\theta})$).
4. Surrogate modeling: Train multi-output GP surrogates to independently model the objective landscape and the feasibility boundary.
5. Acquisition optimization (FIA): Maximize FIA function. This approach blends EI with the probability of feasibility ($P(0 \leq g(\boldsymbol{\theta}) \leq g_{\text{threshold}})$) to propose candidates $\boldsymbol{\theta}_{n+1}$ that are both high-performing and biologically plausible.
6. Two-phase iteration: If insufficient feasible points are found, the acquisition focuses solely on feasibility (Phase A). Once a feasible region is identified, the loop transitions to constrained optimization (Phase B).
7. Convergence: Terminate the process after a fixed number of iterations or when the acquisition value reaches a plateau, indicating the global optimum within the feasible region has been identified.

3. Results

3.1. Initialization and first BO iteration

We followed the steps outlined in Section 2.9 and present the results for the first BO iteration here. Following the methodology in with no elongation rule, we utilized a CC3D agent-based model to simulate cell invasion, where the cell-cell adhesion energy (J_{CC}) and chemoattractant secretion rate (k) were systematically varied. The initialization dataset was generated using a combination of fixed-grid sampling covering $J_{CC} \in [1, 30]$ and $k \in [0, 0.1]$ obtained from preliminary result and twenty Latin hypercube samples of $J_{CC} \in [1, 50]$ and $k \in [0, 0.1]$, to ensure broad coverage of the parameter space. After executing these initial simulations, the objective and constraint functions were calculated for all training points to construct the first surrogate model. The combination of the fixed grid and twenty Latin hypercube samples comprised the initialization data sets for training the GP to obtain a feasible region for our parameter search. Then, we ran our BO analysis to generate the first iteration results (Figures 2 and 3) for calibrating to spheroid and network data, respectively. The plots show the 2D parameter space of $\boldsymbol{\theta} = \{J_{CC}, k\}$ for the GP, the feasible and infeasible regions, the FIA acquisition function, and the EI.

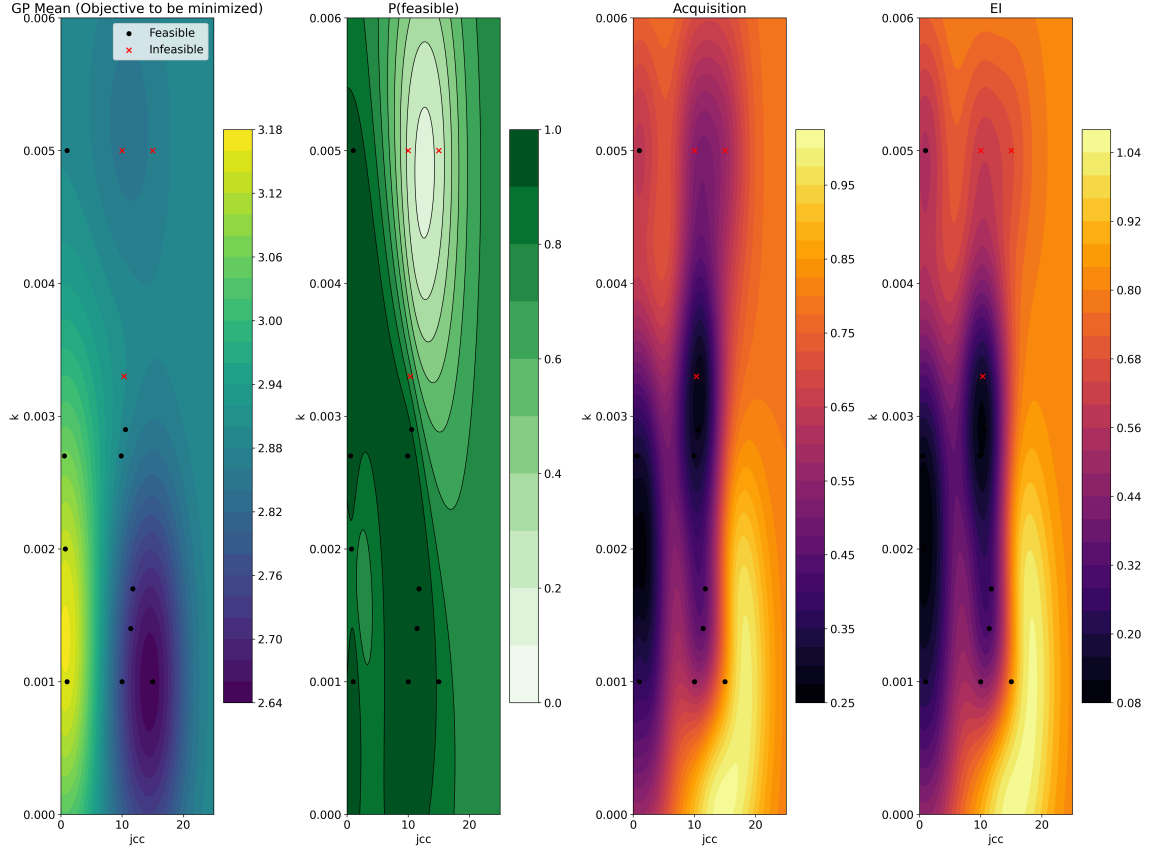


Figure 2: Initial two-dimensional posterior landscapes and acquisition mapping across parameters J_{CC} and k for the spheroid phenotype. (First panel) The Gaussian process (GP) posterior mean provides a preliminary estimation of the WRMSE landscape. (Second panel) The probability of the feasibility map identifies early candidate regions that satisfy the circularity constraint ($0 \leq g(\theta) \leq g_{\text{threshold}}$). (Third panel) The feasibility-informed acquisition (FIA) surface is a weighted average of probability of feasibility and expected improvement (EI). (Fourth panel) EI identifies regions with the largest potential for objective optimization or reduction in uncertainty.

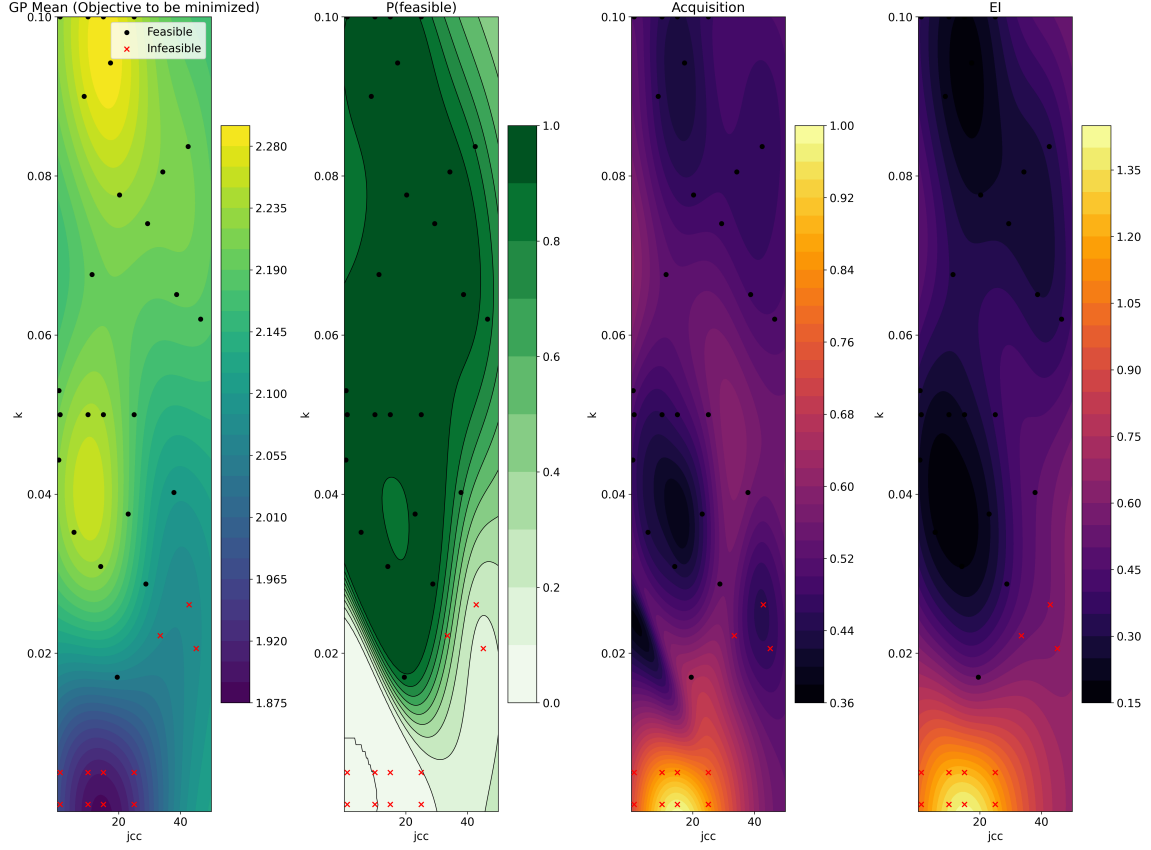


Figure 3: Initial two-dimensional posterior landscapes and acquisition mapping across parameters J_{CC} and k for the network phenotype. (First panel) The Gaussian process (GP) posterior mean provides a preliminary estimation of the WRMSE landscape. (Second panel) The probability of the feasibility map identifies early candidate regions that satisfy the circularity constraint ($0 \leq g(\theta) \leq g_{\text{threshold}}$). (Third panel) The feasibility-informed acquisition (FIA) surface is a weighted average of probability of feasibility and expected improvement (EI). (Fourth panel) EI identifies regions with the largest potential for objective optimization or reduction in uncertainty.

Figures 4 and 5 illustrate the initial one-dimensional slices through the GP and corresponding FIA acquisition functions and feasibility, highlighting the high initial uncertainty and broad exploratory behavior of the Bayesian optimization framework at the onset of calibration, probing the parameter space to identify regions that satisfy circularity-based feasibility constraints for both spheroid and network phenotypes.

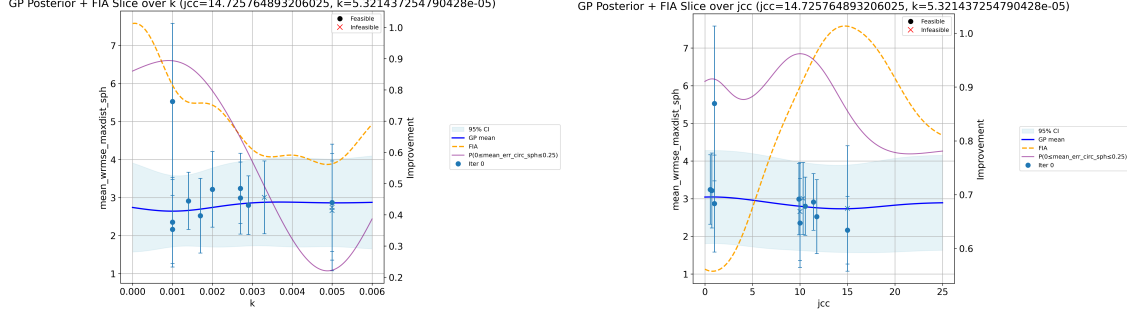


Figure 4: One-dimensional slices of Gaussian process (GP) posterior for the spheroid phenotype at the start of the calibration process (iteration 1) across (left) the invasion rate k and (right) cell-cell adhesion energy J_{CC} . The blue shaded regions represent the 95% confidence interval. The feasibility-informed acquisition (FIA) (orange dashed curve) is a weighted average of expected improvement and probability P of feasibility (dots denote feasible points and x's denote infeasible points).

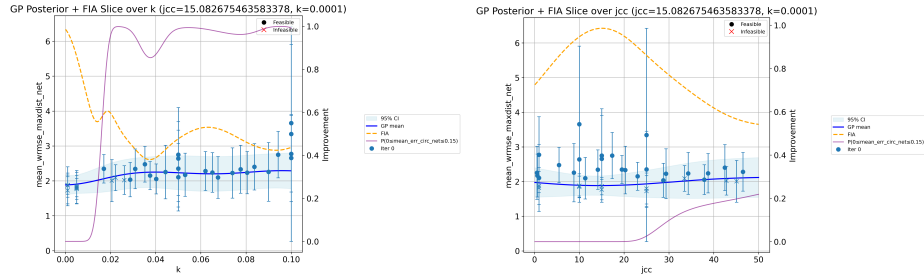


Figure 5: One-dimensional slices of Gaussian process (GP) posterior for the network phenotype at the start of the calibration process (iteration 1) across (left) the invasion rate k and (right) cell-cell adhesion energy J_{CC} . The blue shaded regions represent the 95% confidence interval. The feasibility-informed acquisition (FIA) (orange dashed curve) is a weighted average of expected improvement and probability P of feasibility (dots denote feasible points and x's denote infeasible points).

Figures 6 and 7 present the initial three-dimensional EI and FIA, illustrating the broad exploratory behavior of the optimization framework and its early emphasis on identifying feasible regions that satisfy phenotype-specific morphological constraints.

Figure 8 presents the initial GP surrogate objective landscapes for the spheroid and network phenotypes at the start of the BO process, illustrating the posterior mean WRMSE surfaces and associated 95% confidence intervals across the parameter search space. The predicted WRMSE surfaces are relatively flat and are accompanied by wide confidence

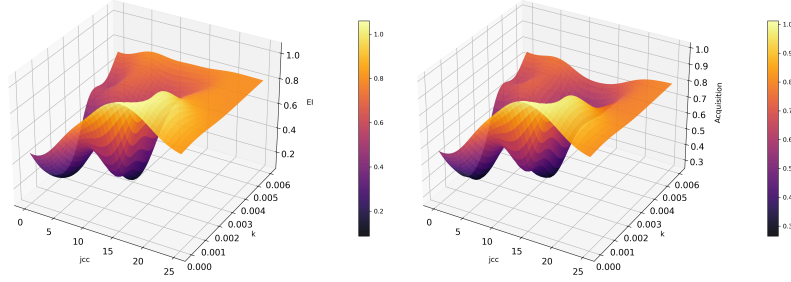


Figure 6: Initial three-dimensional acquisition landscapes for the spheroid phenotype (iteration 1): (left) the expected improvement (EI) surface and (right) the feasibility-informed acquisition (FIA) surface.

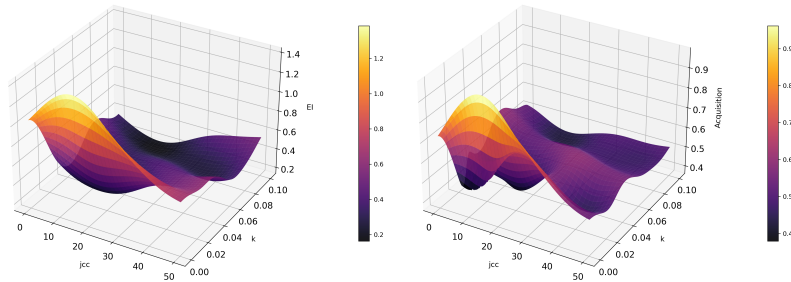


Figure 7: Initial three-dimensional acquisition landscapes for the network phenotype (iteration 1): (left) the expected improvement (EI) surface and (right) the feasibility-informed acquisition (FIA) surface.

intervals, reflecting limited information available to resolve parameter sensitivities at the onset of optimization and the inferred noise from the stochastic simulations. These initial landscapes highlight the high predictive uncertainty and sparse sampling conditions during early optimization, prior to the iterative refinement of biologically feasible and experimentally optimal parameter regions.

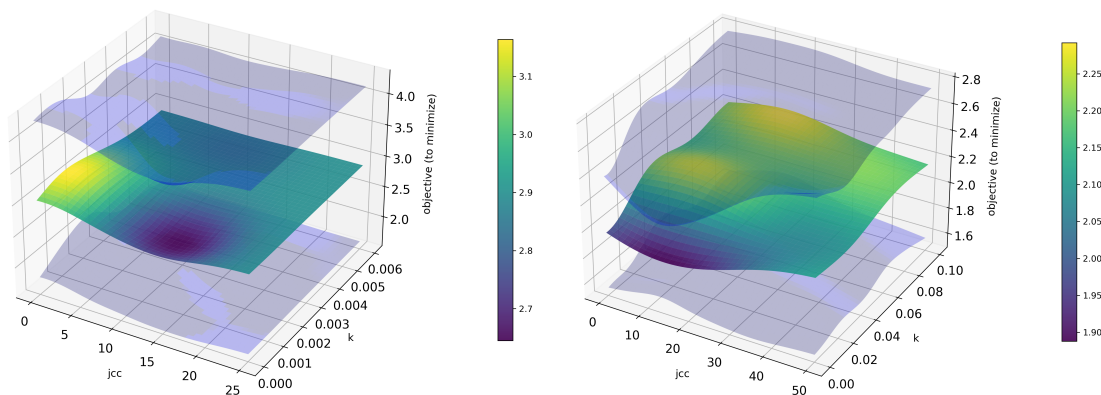


Figure 8: Initial three-dimensional objective landscapes (iteration 1). The surface corresponding to the colorbar represents the GP posterior mean for (left) spheroid and (right) network phenotypes, where the surrounding blue translucent surfaces indicate the 95% confidence intervals. Note the differences in parameter ranges for the two phenotypes.

3.2. Optimized parameters

The final convergent results are shown in Figures 9 and 10. In Figure 9, the optimal configuration (gold star: $J_{CC} = 10.00$, $k = 0.001$) recapitulates the compact, low-invasion morphology observed in 3D culture for the spheroid. The narrow feasible region in the probability of feasibility (compare the reduced domain boundaries in k in Figure 9 to the larger domain in Figure 3) underscores the high sensitivity of spheroid circularity to changes in the secretion rate k , demonstrating the model’s ability to distinguish the spheroid-like behavior from more invasive phenotypes. Through a truncated search space analysis, the GP surrogate models the transitions between collective cell mass stability and fragmented invasion. In Figure 10, the optimization identifies a global minimum (gold star: $J_{CC} = 35.00$, $k = 0.0250$) where the emergence of elongated, invasive filaments is balanced by structural cohesive stability for the network phenotype.

The Pareto analysis in Figure 11 identified a set of non-dominated solutions that balanced minimizing invasion distance error (WRMSE) and circularity error, with the selected optimal configuration (red star) corresponding to the minimum WRMSE within the feasible circularity threshold region. The Pareto front revealed the trade-off between invasion accuracy and morphological agreement, where the chosen optimal parameter set achieved the

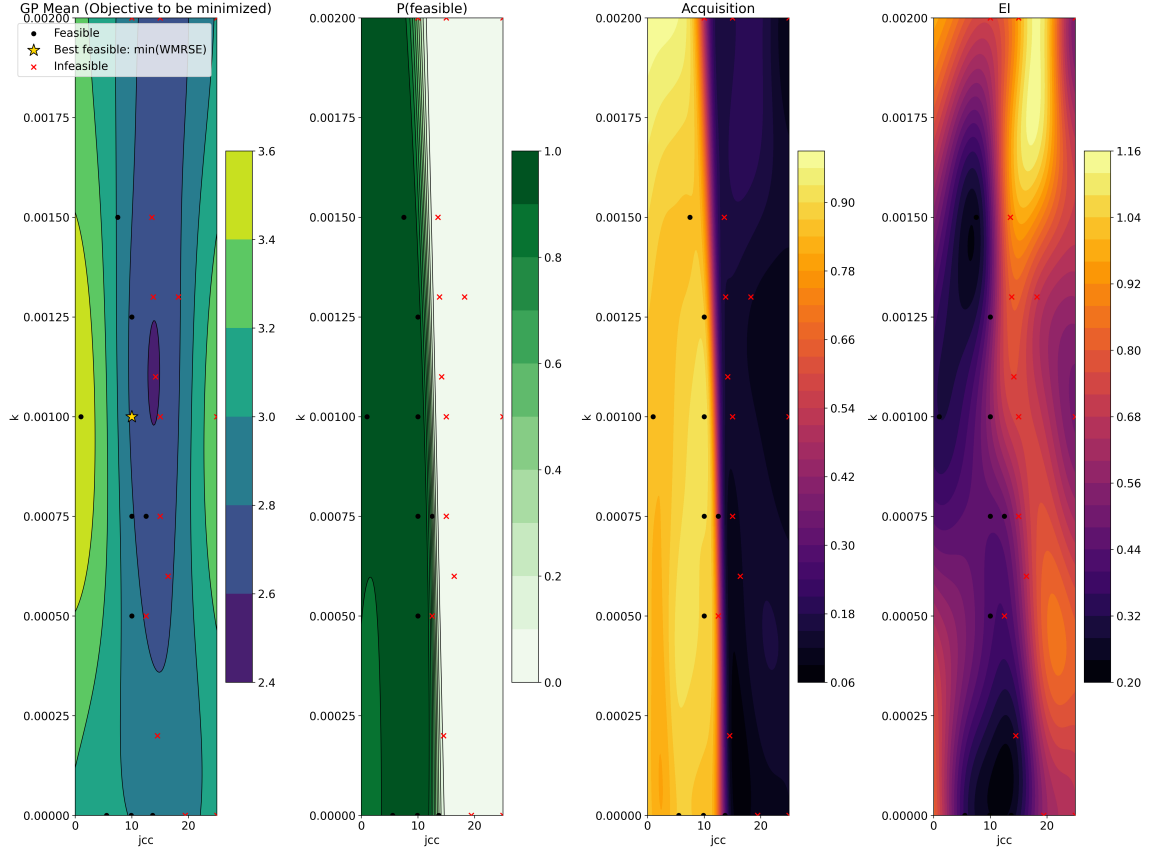


Figure 9: Identification of optimal parameter regimes for the spheroid phenotype using constrained Bayesian optimization. (First panel) The two-dimensional Gaussian process (GP) posterior mean landscape mapping of cell-cell adhesion energy (J_{CC}) and chemoattractant secretion rate (k) to the weighted root mean square error (WRMSE). The gold star denotes the minimum observed error for a feasible value. (Second panel) The probability of the feasibility map distinguishes between feasible and infeasible regions that satisfy the circularity constraint ($0 \leq g(\theta) \leq g_{\text{threshold}}$), with red markers indicating parameter values that violate circularity constraints. (Third panel) The feasibility-informed acquisition (FIA) surface. (Fourth panel) The expected improvement (EI) surface.

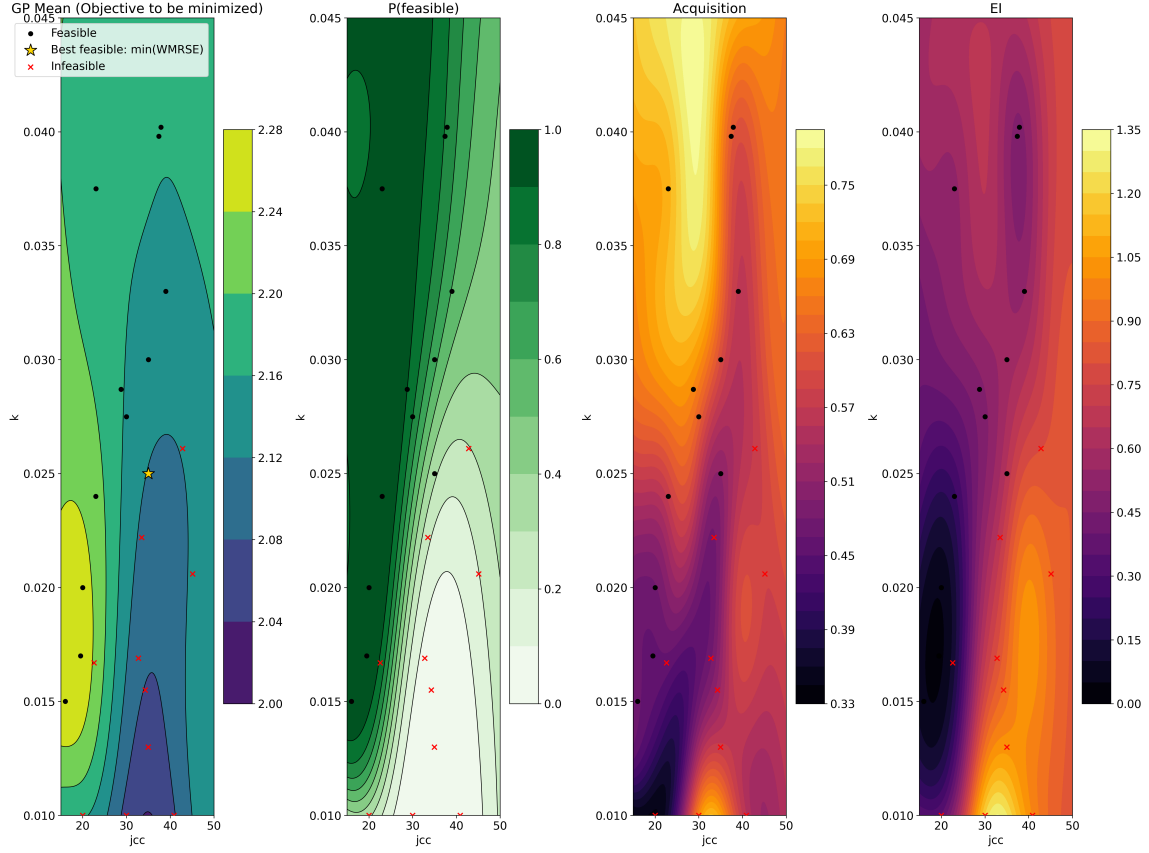


Figure 10: Identification of optimal parameter regimes for the network phenotype using constrained Bayesian optimization. (First panel) The two-dimensional Gaussian process (GP) posterior mean landscape mapping of cell-cell adhesion energy (J_{CC}) and chemoattractant secretion rate (k) to the weighted root mean square error (WRMSE). The gold star denotes the minimum observed error for a feasible value. (Second panel) The probability of the feasibility map distinguishes between feasible and infeasible regions that satisfy the circularity constraint ($0 \leq g(\theta) \leq g_{\text{threshold}}$), with red markers indicating parameter values that violate circularity constraints. (Third panel) The feasibility-informed acquisition (FIA) surface. (Fourth panel) The expected improvement (EI) surface.

lowest feasible WRMSE while maintaining circularity error within the prescribed biological constraint.

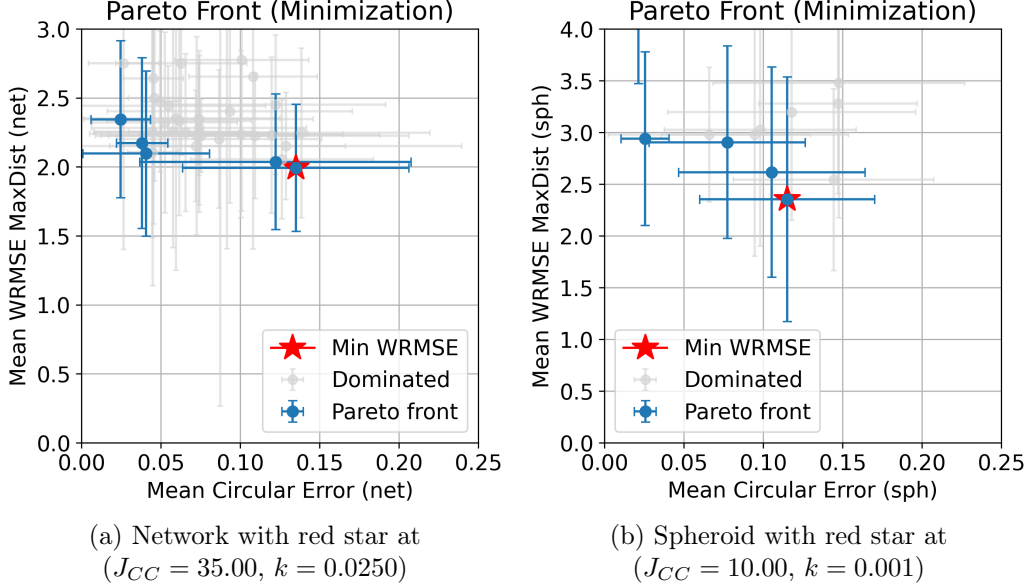


Figure 11: Pareto analysis and error standard deviations for both (left) network and (right) spheroid phenotype optimization. The identified Pareto front (blue circles) represents the trade-off between minimizing invasion distance error (WRMSE) and minimizing circularity error. Points designated as “Dominated” (gray) indicate configurations where improvement in one metric is possible without sacrificing the other. The red star represents the minimum WRMSE within the feasibility threshold.

The final optimized parameters (Table 1) obtained via BO captured the key features of experimentally observed invasion phenotypes. Simulations performed at these optimal parameter values reproduced the spheroid-to-network transition observed in *in vitro* experiments, both qualitatively in terms of morphology and quantitatively in terms of cell invasion and circularity. The resulting invasion networks displayed realistic protrusions strongly validating the mechanistic assumptions embedded in the model.

Table 1: Optimal parameters for network and spheroid phenotypes

Phenotype	Cell-cell adhesion (J_{CC})	Chemoattractant secretion rate (k)	Minimum WRMSE	Circularity error
Network	35.00	0.025	1.99 ± 0.46	0.14 ± 0.07
Spheroid	10.00	0.001	2.35 ± 1.18	0.12 ± 0.06

3.3. Mapping invasion phenotypes across parameter space

The GP surrogate model also enabled systematic exploration of invasion behavior across the parameter space. By evaluating the surrogate prediction across combinations of adhesion and chemotactic parameters, distinct phenotypic regions were identified. Three primary regimes were observed:

- collective spheroid: strong adhesion and minimal invasion,
- dispersed phenotype: very weak adhesion with individual cell migration,
- collective network: intermediate adhesion and moderate chemotaxis.

The collective invasion phenotypes corresponded closely with experimentally observed morphologies.

Figures 12 and 13 show one-dimensional GP posterior slices for the spheroid and network phenotypes, highlighting the sensitivity of the objective landscape to variations in chemoattractant secretion rate (k) and cell-cell junction energy (J_{CC}). These figures further illustrate the convergence behavior of the BO framework through the GP posterior mean, 95% confidence intervals, EI, and feasibility probability, identifying distinct optimal parameter regimes for cohesive spheroid and invasive network phenotypes.

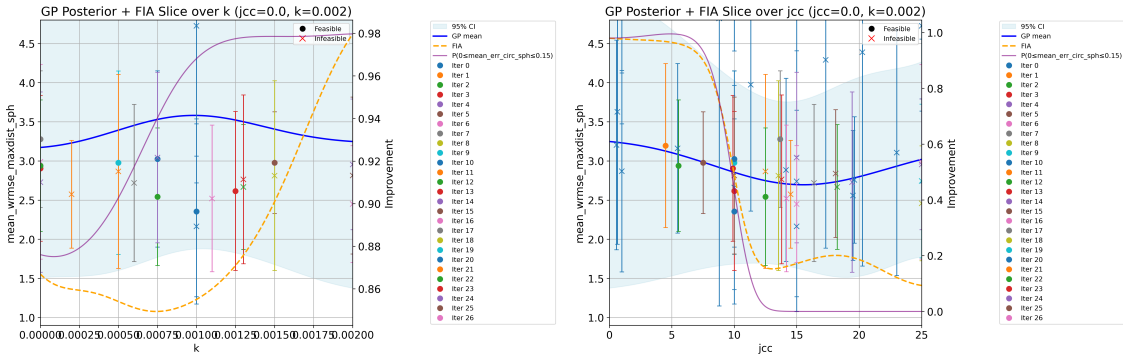


Figure 12: One-dimensional slices of Gaussian process (GP) posterior for the spheroid phenotype at the converged state (iteration 27) across (left) the invasion rate k and (right) cell-cell adhesion energy J_{CC} . The blue shaded regions represent the 95% confidence interval. The feasibility-informed acquisition (FIA) (orange dashed curve) is a weighted average of expected improvement and probability P of feasibility (dots denote feasible points and x's denote infeasible points).

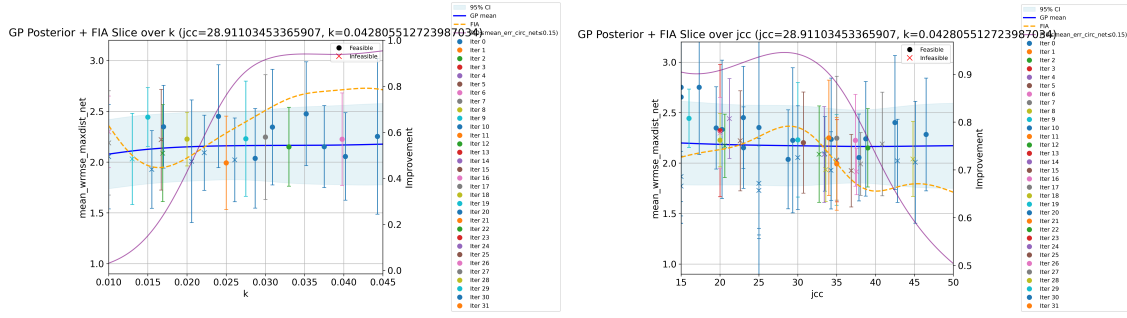


Figure 13: One-dimensional slices of Gaussian process posterior for the invasive network phenotype at the converged state (iteration 32) across (left) the invasion rate k and (right) cell-cell adhesion energy J_{CC} . The blue shaded regions represent the 95% confidence interval. The feasibility-informed acquisition (FIA) (orange dashed curve) is a weighted average of expected improvement and probability P of feasibility (dots denote feasible points and x's denote infeasible points).

Figures 14 and 15 present the final EI and FIA landscapes for the spheroid and network phenotypes, illustrating how the BO framework guided parameter exploration toward biologically feasible and experimentally optimal regions. The acquisition landscapes reveal distinct optimal parameter regimes for each phenotype, where the spheroid phenotype converged toward low chemoattractant secretion rates to preserve cohesive morphology, while the network phenotype favored higher motility and adhesion balance to reproduce invasive branching behavior. The convergence toward low k values ($k = 0.0008$) highlights the restricted parameter regime required to maintain cohesive spheroid morphology. Unlike the spheroid model, the network acquisition landscape favors a higher invasion regime ($k = 0.0250$), identifying the transition point where cell motility and junctional adhesion ($J_{CC} = 35.00$) optimize the match to experimental invasion profiles.

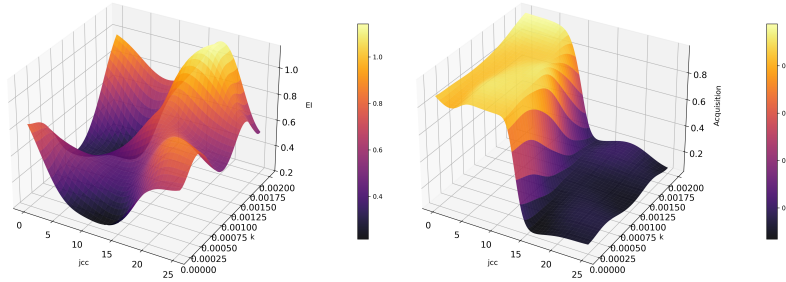


Figure 14: Final three-dimensional acquisition landscapes for the spheroid phenotype (iteration 27): (left) the expected improvement (EI) surface and (right) the feasibility-informed acquisition (FIA) surface.

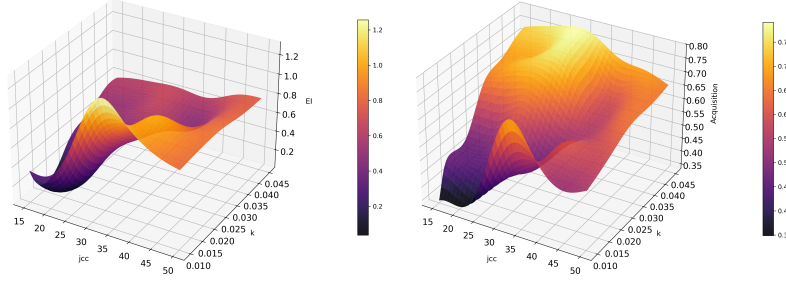


Figure 15: Final three-dimensional acquisition landscapes for the network phenotype (iteration 32): (left) the expected improvement (EI) surface and (right) the feasibility-informed acquisition (FIA) surface.

Figure 16 illustrates the final GP surrogate objective landscapes for the spheroid and network phenotypes, respectively, showing the posterior mean WRMSE surfaces and associated 95% confidence intervals. These figures demonstrate the distinct parameter sensitivities governing each phenotype, where the spheroid phenotype exhibits sensitivity to increased chemoattractant secretion rates, while the network phenotype favors higher cell-cell junction stability to sustain invasive branching structures. The GP posterior surface illustrates the relationship between J_{CC} and k relative to the experimental spheroid WRMSE. The optimal configuration ($J_{CC} = 10.00$, $k = 0.001$) is situated within a sharply defined feasible region, where the rapid increase in WRMSE at higher k demonstrates high sensitivity to motility parameters (Figure 16 left panel). For the network phenotype (Figure 16 right panel), the GP highlights the identified global minimum ($J_{CC} = 35.00$, $k = 0.0250$). The topology reveals a broad fitness valley at higher J_{CC} values, suggesting that networked morphologies require adhesion stability to sustain invasive branching.

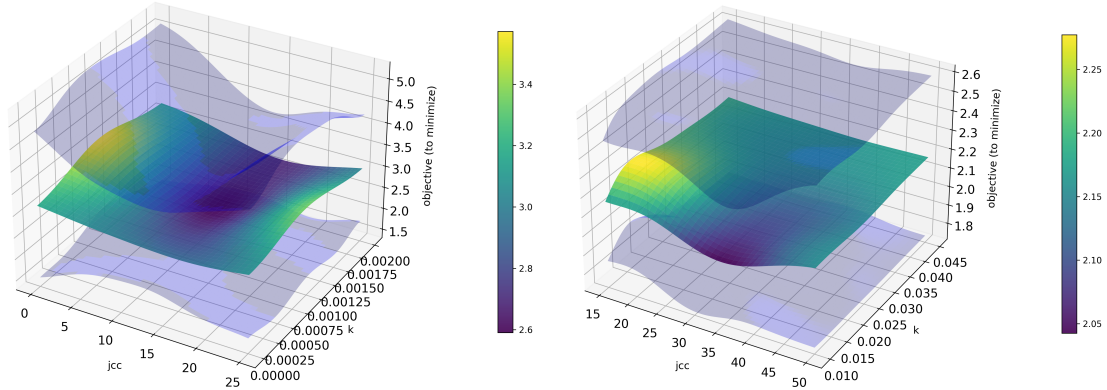


Figure 16: Final three-dimensional acquisition landscapes for spheroid and network phenotypes. The surface corresponding to the colorbar represents the GP posterior mean for (left) spheroid (iteration 27) and (right) network (iteration 32) phenotypes, where the surrounding blue translucent surfaces indicate the 95% confidence intervals. Note the differences in parameter ranges for the two phenotypes.

4. Discussion and conclusions

The data-driven constrained BO framework developed in this study provides a quantitative approach for linking ABM simulations of collective cancer invasion with experimentally observed invasion morphologies. By integrating experimental measurements with *in silico* simulations implemented in CC3D, the framework enables systematic exploration of how biophysical and biochemical parameters, including cell-cell adhesion and chemoattractant secretion rate, interact to generate complex invasion phenotypes. A central experimental observable used in this work is cell circularity, which reflects morphological transformations associated with cancer progression. During epithelial-mesenchymal transition (EMT), epithelial cells lose their characteristic rounded morphology and adopt elongated shapes associated with enhanced migratory capacity and metastatic potential. Reduced circularity, therefore, serves as a measurable indicator of invasive phenotypes. Incorporating circularity as a feasible constraint alongside invasion distance allowed the calibration procedure to capture both morphological and spatial characteristics of collective invasion.

Our final convergent results from Figures 9 and 10 indicate that collective invasion emerges from a narrow region of parameter space in which cell-cell adhesion and chemotactic secretion rate are balanced. When cell adhesion energy is strong, cells remain confined within compact spheroids and fail to invade the surrounding space. Conversely, excessively weak cell adhesion energy leads to cell dispersal and loss of coordinated movement. Optimal invasion occurs under intermediate cell adhesion levels that permit partial detachment while preserving intercellular cohesion, enabling coordinated streaming of cells during invasion. Similarly, optimized chemoattractant secretion rates generate self-reinforcing gradients that stabilize invasion fronts. Sensitivity analysis further revealed that secretion levels exceeding

a critical threshold destabilize these fronts, producing branching invasion patterns reminiscent of diffusion-limited growth observed in invasive tumors. These findings support the hypothesis that invasion plasticity arises from coupled physical and biochemical feedback mechanisms rather than from regulation by a single pathway. In the model, cells with moderate chemoattractant secretion produce network-like invasion morphologies. The resulting spheroid-to-network transitions are consistent with experimental observations of phenotypic plasticity in collective cancer invasion. The calibrated parameter regimes identified here align with experimental and computational studies of collective invasion. Experiments in three-dimensional collagen and Matrigel environments have reported similar transitions between non-invasive spheroids and invasive networks under varying extracellular matrix stiffness and chemokine concentrations. Studies by Friedl et al.²⁵ and Haeger et al.²⁶ emphasize partial loss of cell-cell adhesion, cytoskeletal polarization, and chemotactic signaling as defining features of collective invasion. These features are reproduced in the parameter regimes identified by the Bayesian calibration process. Computational studies using CPMs, including those by Szabó et al.²⁷, have reported comparable relationships between adhesion and motility parameters. However, earlier studies typically relied on heuristic parameter tuning. In contrast, the BO strategy presented here provides a reproducible and data-informed calibration pipeline that quantitatively aligns model outputs with experimental observations.

Beyond its biological implications, the framework introduces several methodological advances for parameter estimation in complex ABMs. Surrogate-assisted constrained BO significantly improves computational efficiency by reducing the number of simulations required to explore multidimensional parameter spaces. GP surrogate models guide the search toward promising parameter regions while expected improvement balances exploration and exploitation during optimization. The Bayesian formulation also provides posterior distributions for model parameters, enabling explicit quantification of uncertainty and identification of parameters that are weakly constrained by available data. These features improve both the transparency and reproducibility of model calibration in biological simulations. Validation against independent experimental datasets confirmed that the calibrated model reproduces key descriptors of invasion morphology. Simulated invasion distances closely matched experimental measurements, while predicted circularity and branching characteristics were consistent with observed invasion structures. The model also captured the temporal progression of invasion, transitioning from compact clusters during early simulation stages to extended network morphologies at later stages, consistent with time-lapse imaging observations. Together, these results demonstrate that the framework captures both the structural and dynamic aspects of collective invasion. More broadly, this work illustrates how data-driven and probabilistic modeling strategies can transform mechanistic biological modeling into a predictive scientific framework. By integrating experimental observations directly into the calibration process and explicitly accounting for uncertainty, computational models can move beyond qualitative hypothesis testing toward quantitative prediction of biological behavior.

BO provided several advantages for the calibration problem by achieving computational efficiency through requiring fewer simulations to reach convergence, thereby making computationally intensive models tractable. Its probabilistic surrogate improved statistical robustness by mitigating noise and stochastic fluctuations inherent in CC3D simulations, while the exploration–exploitation trade-off prevented entrapment in local minima by systematically probing uncertain regions and prioritizing promising parameter regimes. In addition, the GP surrogate enhanced interpretability by revealing how parameter variations influenced model outcomes, including nonlinear dependencies and synergistic effects, and the framework remained scalable, allowing straightforward extension to multidimensional settings or multi-objective optimization tasks such as jointly minimizing WRMSE and computational runtime.

In summary, BO effectively replaced heuristic parameter tuning with a data-informed, probabilistic, and automated search that substantially improved both accuracy and efficiency in modeling collective cancer cell invasion. By integrating surrogate modeling, probabilistic reasoning, and biological constraints, BO provided a powerful bridge between experimental data and mechanistic simulation. The following section explores how data-driven and Bayesian approaches can be combined-leveraging the strengths of both to achieve hybrid optimization strategies capable of addressing even more complex biological systems. In this study, we developed a unified calibration framework that integrates Bayesian parameter estimation with surrogate-assisted optimization to quantitatively calibrate ABMs of collective cancer invasion. By combining experimental image-derived metrics with mechanistic simulations in CC3D, the framework transforms parameter calibration from a heuristic process into a systematic and reproducible methodology. The calibrated model reveals that invasion phenotypes emerge from a coupled balance between cell–cell adhesion, cell elongation, and chemotactic signaling. These interacting mechanisms regulate transitions between compact spheroidal clusters and network-like invasion patterns, providing mechanistic insight into the physical processes underlying metastatic plasticity. Beyond the specific application to cancer invasion, the methodology provides a generalizable strategy for calibrating complex biological simulations using experimental data. Future work may extend this framework to multi-objective BO, incorporate more advanced surrogate models such as physics-informed neural networks, and develop hierarchical inference strategies that couple cell-level dynamics with tissue-scale invasion patterns. Ultimately, such approaches may enable predictive, data-calibrated computational models that serve as digital analogs of biological systems and support model-driven discovery in cancer systems biology and related fields.

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